Electricity & Magnetism

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1 Electric Charges and Fields

1.1 Electric Charge and Electric Force

Electric Charge

- **Electric charge** (q or Q) is a fundamental property of matter.
- Electric charge is measured in the SI derived unit *Coulombs* (C).
  - Coulombs are equal to *ampere seconds*, being the amount of charge transferred in a 1 A current in one second ($\mathrm{C} = \mathrm{A} \cdot \mathrm{s}$).
- Electric charge comes from **charge carriers**: protons (p^+) and electrons (e^-)
  - The magnitude of charge of both protons and electrons are equal to the *elementary charge*:

$$\text{Elementary Charge} = e = 1.6 \cdot 10^{-19}\ \mathrm{C}$$

$$|q_{\text{proton}}| = |q_{\text{electron}}| = e.$$

  - Protons p^+ carry a positive charge of $+e$, and electrons e^- carry a negative charge of $-e$.
- Electric charge is a **quantized scalar**.
  - **Quantized:** electric charge can only be integer multiples of the elementary charge e.
    In other words, something can only have a charge of $q = ke$, where $k \in \mathbb{Z}$ and e is the elementary charge.
  - **Scalar:** charge does not have direction, but it can be either positive or negative.

Example:
An object consists of 4 protons and 6 electrons. What is its charge in Coulombs?

$$\begin{aligned} Q = (n_{\text{protons}} - n_{\text{electrons}})e = (4-6)e &= -2e \\ &= -2(1.60 \cdot 10^{-19}\ \mathrm{C}) \\ &= -3.20 \cdot 10^{-19}\ \mathrm{C}. \end{aligned}$$

Law of Charges
Two charges of opposing signs *attract*, and two charges of the same sign *repel*.

- A **point charge** refers to an object or system which has charge, but has negligible physical size. Point charges are easy to deal with because the direction from the charge to any object solely depends on the position of the object.

Coulomb's Law

Coulomb's Law

Let q_1 and q_2 be the charges of two point charges a distance r between their centers of charge. Then, the *electrostatic force* $\vec{F}_e$ exerted by q_2 on q_1 is:

$$\text{Electrostatic Force} = \vec{F}_e = k_e \frac{q_1 q_2}{r^2} \hat{r}$$

where:

- k_e is **Coulomb's constant**, defined as $k_e = \frac{1}{4\pi\varepsilon_0} \approx 8.99 \cdot 10^9\ \mathrm{N\,m^2\,C^{-2}}$
  - ε_0 is the **permittivity of free space**, which is a constant which we will define later.
- $\hat{r}$ is the unit vector directed *away* from the location of q_1.
  - in other words, if q_1 and q_2 are the same sign, the force will repel the charges *away* from each other.

- By Newton's Third Law, $\vec{F}_{e_{1,2}} = -\vec{F}_{e_{2,1}}$. Two charges will always exert a pair of electrostatic forces on each other.
- Notice that Coulomb's Law mirrors the Universal Law of Gravitation, $\vec{F}_G = G\frac{m_1 m_2}{r^2}\hat{r}$.
  - $G \approx 6.67 \cdot 10^{-11}\ \mathrm{N\,m^2\,kg^{-2}}$, the gravitational constant, is *much* smaller than k_e. This means that for large scale objects which are *charged*, gravitational force is negligible compared to the electrostatic force.

    However, it is often the case that large objects *aren't* charged (they are neutral since), which is why we still see the effects of gravitational force.
  - In the case of $\vec{F}_G$, $\hat{r}$ points *towards* the other object rather than *away*. Gravity is always an attractive force. In contrast, electrostatic force can be either attractive or repulsive.

Law of Superposition for Coulomb's Law (Discrete)

Let $q_1, q_2, \ldots, q_n$ be a discrete set of charges in space. A particle of charge Q will have its electrostatic force be the vector sum of all individual electrostatic forces:

$$\vec{F}_e = \sum_i \vec{F}_{e_{i,Q}} = \sum_i k_e \frac{q_i Q}{r^2} \hat{r}.$$

In other words, the net electrostatic force is the sum of all individual electrostatic forces.

We can generalize this law to continuous charge distributions. For every small change in charge along a charged object dq, the change in electrostatic force $d\vec{F}_e$ is:

$$d\vec{F}_e = k_e \frac{Q\, dq}{r^2} \hat{r}.$$

Thus, by the law of superposition:

Law of Superposition for Coulomb's Law (Continuous)

Let Q be a point charge and let an object have a continuous charge distribution $\mathcal{D}$. Then, the net electrostatic force $\vec{F}_e$ must be:

$$\vec{F}_e = k_e \int_{\mathcal{D}} \frac{Q}{r^2} \hat{r} \, dq.$$

1.2 Conservation of Electric Charge

Conservation of Charge

The Law of Conservation of Charge

For isolated systems, net charge is *conserved*:

$$\Sigma Q_i = \Sigma Q_f.$$

This law holds because in an isolated system, charge carriers (i.e. electrons) cannot enter or leave the system.

- Charge cannot be created or destroyed, only *transferred* between objects or particles.
- Applies at all scales: subatomic (e.g., particle interactions), macroscopic (e.g., charging a balloon), and circuit-level.

Methods of Charging

- An object can have a change in charge when they are introduced into a new system, allowing the charge of the system to distribute into the object.
  - In almost all cases, only electrons freely move and cause charge transfers.
- When a previously neutral object obtains charge, we say that the object has been *charged*.

Insulators and Conductors

- It is difficult to change the charge of an **insulator**. Electrons are bound tightly to the atomic nuclei, and cannot easily move. (Electrons are *localized*).
  - We say that the electrons *resist* movement.
  - The degree to which the electrons of a material resist movement is called *resistivity*.
- It is easy to change the charge of a **conductor**. Electrons can freely move (they are *delocalized*).
  - The degree to which the electrons of a material are able to move is called *conductivity*. It is the inverse of resistivity (conductivity = $\frac{1}{\text{resistivity}}$).
- **Semiconductors** have properties resembling both conductors and insulators. Electrons are somewhat bound to the atomic nuclei, and conductivity *varies* based on the composition of materials.

Charging by Conduction (Contact)

- When an object touches a *conductor* charge can be directly transferred through physical contact. The total charge of the object-conductor system is conserved.
- The distribution of charges between the individual objects depend *only* on their **initial charges** and their **sizes (charge capacities)**.

- Two *identical* objects of this manner will evenly distribute charge.
  ex: sphere A (+4 μC) and sphere B (0 μC) make contact with each other.
  If spheres A and B are identical in size, then the final charge on each is +2 μC.
  - To generalize, the final charge of each object after the *conduction* of n identical objects of initial charges $q_1, q_2, \ldots, q_n$ is simply the average charge:

$$q_{\text{individual object after conduction}} = \frac{q_1 + q_2 + \cdots + q_n}{n}.$$

Charging by Induction (Contactless)

- A nearby conductor may *induce* a redistribution of charge in an object. We call this process **induction**.

1.3 Electric Fields

Electric Field

Let $\vec{E}$ be the net **electric field** at any point P. Then, if a particle with charge q is placed at point P, then the net electrostatic force on the particle is:

$$\vec{F}_e = q\vec{E}.$$

By Coulomb's Law, the electric field at a point P induced by a point charge Q is:

$$\vec{E}_P = \frac{\vec{F}_e}{q} = k_e \frac{Q}{\mathrm{r}^2}\hat{\mathrm{r}}$$

where $\vec{\mathrm{r}}$ is the distance vector from the position of Q to P.

- The **electric field** is the *electrostatic force per unit charge*.
  - Take a test charge q and place it a distance r from another charge Q. The charge Q induces an electrostatic force $\vec{F}_e$ on q and the electric field from Q is $\vec{F}_e/q$.
- The electric field is a vector function of position ($\vec{E}(x, y, z)$), making it a vector field
  (the electric field at every point is a vector).
- Electric field has units of *newtons per coulomb*, N / C
  - $\mathrm{N\,C^{-1}} = \mathrm{kg\,m\,s^{-3}\,A^{-1}}$ in SI base units.

Electric Field Lines

- Electric field lines generalize the electric field to general regions in space.
- Electric field lines will always move away from positive charges and into negative charges.
- Electric field lines will never cross, and by convention, lines drawn closer together indicate a greater magnitude of electric field.

1.4 Electric Fields of Charge Distributions

We can apply the law of superposition for electrostatic force to electric fields.

Discrete Charge Distributions

For discrete charge distributions (a finite set of point charges):

$$\vec{E} = \frac{\vec{F}_e}{Q} = \frac{\sum_i \vec{F}_{e_{i,q}}}{\sum_i q_i} = \sum_i \vec{E}_i.$$

In other words, the net electric field at a given position is the sum of the electric fields induced by all individual point charges.

Continuous Charge Distributions

Often times, we will deal with objects with non-negligible sizes where we must consider the **distribution of charge** throughout the object.

If we split an object into many small, negligibly-sized portions, we can consider each small portion its own point charge that contributes a small amount towards the total electric field.

Let dq be the additional charge of one of these negligibly-sized point charges. Then the additional electric field $d\vec{E}$ induced by this point charge is:

$$d\vec{E} = k_e \frac{dq}{r^2}\hat{r}.$$

The law of superposition allows us integrate both sides to obtain the net electric field:

$$\vec{E} = k_e \int \frac{1}{r^2}\hat{r}\, dq.$$

Note

The fact that $\hat{r}$ appears inside our integral can make calculations difficult, since the direction of each $\hat{r}$ can vary as we integrate through different positions along our object.

Luckily, at this level, we will only consider scenarios where $\hat{r}$ can be considered constant or where there is symmetry.

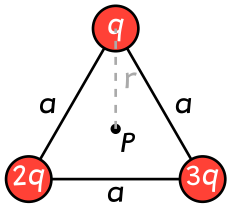
Often times, the charge differential dq is not directly given to us, but rather indirectly using a **charge density**. There are three charge densities:

- **Linear charge density** (λ), charge per unit length.
  - By definition, $\lambda = \frac{dq}{dl} \implies dq = \lambda\, dl$ where dl is a small change in length.
  - SI units are *coulombs per meter*, C/m
- **Surface charge density** (σ), charge per unit surface area.
  - By definition, $\sigma = \frac{dq}{dA} \implies dq = \sigma\, dA$ where dA is a small change in surface area.
  - SI units are *coulombs per square meter*, C/m^2

- **Volumetric charge density** (ρ), charge per unit volume.
  - By definition, $\rho = \frac{dq}{dV} \implies dq = \rho\, dV$ where dV is a small change in volume.
  - SI units are *coulombs per cubic meter*, C/m^3

If charge density is constant, the object is **uniformly charged**.

Example: Finding the Electric Field from Discrete Charges



Three point charges q, $2q$, and $3q$ are arranged in an equilateral triangle of side length a as shown. Find the electric field at the center of the triangle.

The distance r from a vertex to point P is $r = \frac{a}{\sqrt{3}}$. By Coulomb's law, the electric field magnitude due to a charge Q a distance r away is:

$$E = \frac{k_e Q}{r^2} = \frac{k_e Q}{\left(\frac{a}{\sqrt{3}}\right)^2} = \frac{3 k_e Q}{a^2}.$$

where k_e is Coulomb's constant.

Thus, the magnitudes of electric field caused by each charge is:

$$E_1 = \frac{3k_e q}{a^2} \qquad E_2 = \frac{6k_e q}{a^2} \qquad E_3 = \frac{9k_e q}{a^2}.$$

By the law of superposition, $\vec{E}_P = \vec{E}_1 + \vec{E}_2 + \vec{E}_3$. However, we only have magnitudes, so we have to use trigonometry to break each magnitude into basis components.

Recall the electric field points *away* from the charge. For example, at point P, $\vec{E}_1$ would point downwards. This means $\vec{E}_1$ has no x-component and a negative y -component, so we derive $\vec{E}_1 = -E_1 \hat{\jmath}$.

$\vec{E}_2$ makes a 30° angle with the positive x-axis, so $\vec{E}_2 = E_2(\hat{\imath}\cos 30° + \hat{\jmath}\sin 30°)$. Similarly, $\vec{E}_3$ makes a 30° angle with the *negative* x-axis, so $\vec{E}_3 = E_3(-\hat{\imath}\cos 30° + \hat{\jmath}\sin 30°)$.

Writing out the components:

$$\begin{aligned}
\vec{E}_P &= \left(E_2 \cos 30° - E_3 \cos 30°\right)\hat{\imath} + \left(-E_1 + E_2 \sin 30° + E_3 \sin 30°\right)\hat{\jmath} \\
&= \left(E_2 - E_3\right)\frac{\sqrt{3}}{2}\hat{\imath} + \left(-E_1 + \frac{E_2 + E_3}{2}\right)\hat{\jmath} \\
&= \left(-\frac{3k_e q}{a^2}\right)\frac{\sqrt{3}}{2}\hat{\imath} + \left(-3 + \frac{6+9}{2}\right)\frac{k_e q}{a^2}\hat{\jmath} = -\frac{3\sqrt{3}k_e q}{2a^2}\hat{\imath} + \frac{9k_e q}{2a^2}\hat{\jmath}.
\end{aligned}$$

Example: Electric Field from a Continuous Charge Distribution

Point P is a distance b directly to the left of a uniformly charged rod of length L with linear mass density λ. Derive an expression for the electric field at P.

First, since the rod is not a point charge, we are dealing with a *continuous charge distribution*:

$$\vec{E}_P = k_e \int_{x=0}^{x=L} \frac{1}{r^2}\hat{r}\, dq.$$

Let x be how far from the left side of the rod we are (so we integrate from 0 to L).

Realize that the direction of the electric field is always to the left. So, $\hat{r}$ must be constant and equal to $-\hat{\imath}$ (left-pointing unit vector). Also recognize that r is the distance from P to the left side plus x ($r = b + x$):

$$\vec{E}_P = k_e(-\hat{\imath}) \int_{x=0}^{x=L} \frac{1}{(b+x)^2}\, dq$$

Since we are given a linear mass density, use $dq = \lambda\, dx$:

$$\vec{E}_P = -k_e\hat{\imath} \int_0^L \frac{\lambda}{(b+x)^2}\, dx.$$

Since we know that the rod is *uniformly charged*, λ is constant, and we can now solve the integral and simplify:

$$\begin{aligned}\vec{E}_P &= -k_e\lambda\hat{\imath} \int_0^L \frac{1}{(b+x)^2}\, dx \\ &= -k_e\lambda\hat{\imath}\left[-\frac{1}{b+x}\right]_0^L \\ &= k_e\lambda\left(\frac{1}{b+L} - \frac{1}{b}\right)\hat{\imath} \\ &= -\frac{k_e\lambda L}{b(b+L)}\hat{\imath}.\end{aligned}$$

Since total charge $Q = \int_0^L dq = \int_0^L \lambda\, dx = \lambda L$, we can simplify in terms of Q:

$$\vec{E}_P = -\frac{k_e Q}{b(b+L)}\hat{\imath}$$

1.5 Electric Flux

Electric Flux

Let $\vec{E}$ be the electric field and $d\vec{A}$ be a vector element of area on a surface S. The **electric flux** Φ_E through the surface is:

$$\Phi_E = \iint_S \vec{E} \cdot d\vec{A}$$

- $\vec{E}$ is the electric field at each point on the surface.
- $d\vec{A}$ is the vector with its magnitude being a small change in surface area, pointing perpendicularly outwards from the surface
  ($d\vec{A} = \hat{n}\, dA$ where $\hat{n}$ is the unit normal to S)

- **Electric flux** measures how much electric field "passes through" a given surface.
  - If $\vec{E}$ is parallel to $d\vec{A}$, the flux is maximized.
  - If $\vec{E}$ is perpendicular to $d\vec{A}$, the flux is zero.
  - The dot product ensures that only the component of the electric field perpendicular to the surface contributes to flux.
  - The surface here does not need to be an actual physical surface. It can be an imaginary surface which we use to analyze flux with, called a **Gaussian surface**.
- The symbol Φ_E or φ_E is used to refer to electric flux. It is a scalar quantity.
  (or just Φ if no need to clarify which type of flux)
- The preferred unit for electric flux is the *volt-meter*, $\mathrm{V \cdot m}$.
  - The unit derived from the equation is *newton-meter squared per coulomb*, $\mathrm{N\,m^2\,C^{-1}}$.
  - *A volt* (V) is a measure of *electrical potential energy per unit charge* (which will be covered later), so volts have units of $\mathrm{J/C = N\,m\,C^{-1}}$ (unit of energy divided by unit of charge).
  - Thus, a *volt-meter* is equivalent to a *newton-meter squared per coulomb*.
- If electric field is constant everywhere on the surface AND our surface is flat, then our equation simplifies:

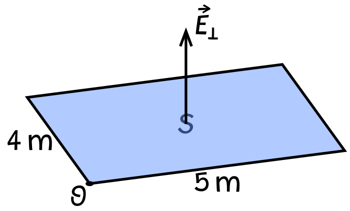
$$\Phi_E = \vec{E} \cdot \iint_S d\vec{A} = \vec{E} \cdot \vec{A}.$$

- By the definition of the dot product, the electric field for flat surfaces is:
  - $\Phi_E = EA\cos\theta$, if electric field is constant
  - $\Phi_E = \iint_S E\cos\theta\, dA$, if electric field is not constant

  E is the magnitude of the electric field, A is surface area, and θ is the angle between the outwards-pointing normal vector to the surface and the electric field vector.

Example:

A non-uniform electric field $\vec{E} = \langle 2xy, z^2, 2x+3y \rangle$ N/C passes through a flat, rectangular surface S as shown in the diagram. What is the electric flux passing through S? Assume x, y, z are specified in meters.



First, since we are only concerned about the component of the electric field that is perpendicular to S, we know that our effective electric field is $E = 2x + 3y$ (i.e. we only consider the z-component of the electric field).

Now, this becomes a simple double integral:

$$\begin{aligned}
\Phi_E &= \iint_S (2x+3y)\,\mathrm{d}A \\
&= \int_{y=0}^{4}\int_{x=0}^{5} (2x+3y)\,\mathrm{d}x\,\mathrm{d}y \\
&= \int_0^4 \left(x^2 + 3xy\right)\Big|_{x=0}^{x=5}\,\mathrm{d}y \\
&= \int_0^4 (25 + 15y)\,\mathrm{d}y \\
&= \left(25y + 7.5y^2\right)\Big|_0^4 = 220 \ \ \mathrm{V \cdot m}.
\end{aligned}$$

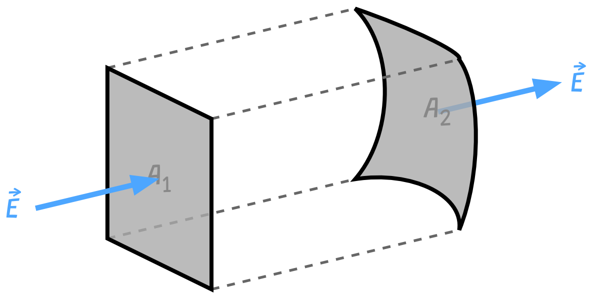
Projection Principle

The *projection principle* states that the electric flux through any curved surface S is equal to the electric flux through the projection of that surface onto a plane perpendicular to the electric field.

If the direction of the electric field is not constant, we can break the curved surface into many small flat surfaces where the electric field is approximately constant, and apply the projection principle to each small surface.

This principle holds because for every small patch of area $\mathrm{d}A$, the contribution to flux is:

$$\mathrm{d}\Phi_E = \vec{E} \cdot \mathrm{d}\vec{A} = E\,\mathrm{d}A\cos\theta = E\,\mathrm{d}A_\perp.$$



The electric flux through A_1 is equal to the electric flux through A_2.

1.6 Gauss' Law

Assume an isolated point charge of magnitude q is situated at the center of a sphere of radius r. What is the total electric flux over the surface S of the sphere?

Using Coulomb's Law, we see that the magnitude of electric field $\vec{E}$ at any point on the sphere (i.e. a distance r from the charge) is:

$$\left|\vec{E}\right| = k_e \frac{q}{r^2}.$$

Also realize that:

- the magnitude electric field is constant for all points on the surface of the sphere
- the electric field $\vec{E}$ points in the same direction as $d\vec{A}$ for all points on the surface

Thus, the formula for electric flux simplifies:

$$\Phi_E = \iint_S \vec{E} \cdot d\vec{A} = \iint_S \left|\vec{E}\right| dA = E \iint_S dA = EA$$

Since A would simply be the surface area of the sphere $A = 4\pi r^2$:

$$\begin{aligned}\Phi_E = EA &= \left(k_e \frac{q}{r^2}\right)\left(4\pi r^2\right) \\ &= \left(\frac{1}{\cancel{4\pi}\varepsilon_0} \frac{q}{\cancel{r^2}}\right)\cancel{4\pi r^2} \\ &= \frac{q}{\varepsilon_0}\end{aligned}$$

It turns out that *all closed surfaces* which enclose a volume ("**Gaussian surfaces**") can be projected as spheres, and thus this formula applies to all such surfaces. Formally:

Gauss' Law (Integral Form)

Let Φ_E be the *electric flux* through a **closed surface** S. Then:

$$\Phi_E = \oint\!\!\!\oint_S \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enclosed}}}{\varepsilon_0}$$

where:

- $\vec{E}$ is the electric field
- $d\vec{A}$ is the area vector differential ($d\vec{A} = \hat{n}\, dA$ where $\hat{n}$ is the unit normal to S)
- Q_{enclosed} is the total charge **inside** the volume enclosed by S
- ε_0 is the permittivity of free space.

- The electric flux through any closed surface is proportional to the net charge enclosed within that surface.
- We can choose any closed surface we want (even imaginary ones) to take advantage of this law.
- A common strategy is to choose a Gaussian surface which takes advantage of symmetry to make calculating the electric field easier.

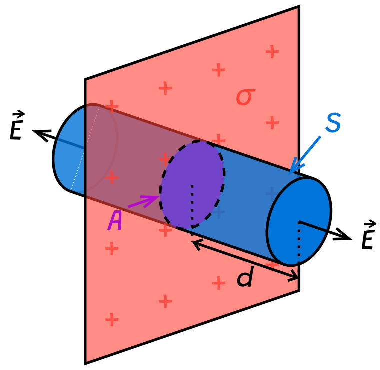
- Where the electric field is perfectly perpendicular to our surface, then it is parallel to $\hat{n}$ and $\vec{E}\cdot d\vec{A} = E\,dA$ (or $-E\,dA$ if the field points inwards).
- Where the electric field is perfectly parallel to our surface, then it is perpendicular to $\hat{n}$ and $\vec{E}\cdot d\vec{A} = 0$.

Example: Using Gauss' Law to find Electric Field

An infinite sheet of charge has a uniform surface charge density σ. Use Gauss' Law to derive an expression for the electric field a distance d from the sheet.

The first step is to choose the Gaussian surface.

- The electric field points directly away from the sheet on both sides
- We can choose a cylindrical surface where the electric field pierces through the two circular ends perpendicularly, and is parallel to the curved side of the cylinder. This way, we only have to consider the flux through the two circular ends of the cylinder.



Let the area of each circular end be A, and have the cylinder extend a distance d on either side of the sheet. We can then let the magnitude of electric field at each end be E.

Let's evaluate the left-hand side of Gauss' Law, the flux integral.

The electric field is parallel to the curved side of the cylinder, so they contribute no flux. At each circular end, the electric field is perpendicular to the surface, so the flux through each end is simply EA. Since there are two ends, the total flux is:

$$\Phi_E = EA + EA = 2EA.$$

For the right-hand side, the total charge enclosed by the cylinder is simply the charge on the portion of the sheet inside the cylinder. Since the sheet has a uniform surface charge density σ, the total charge enclosed is:

$$Q_{\text{enclosed}} = \sigma A.$$

Finally, applying Gauss' Law:

$$\Phi_E = \frac{Q_{\text{enclosed}}}{\varepsilon_0}$$

$$2EA = \frac{\sigma A}{\varepsilon_0}$$

$$E = \frac{\sigma}{2\varepsilon_0}.$$

Introduction to Maxwell's Equations

Gauss' Law is typically the first equation introduced in the set of four equations fundamental to electromagnetism known as *Maxwell's Equations*:

Maxwell's Equations

Name	Integral Form	Differential Form
Gauss' Law	$\oint\!\!\oint_S \vec{E}\cdot d\vec{A} = \frac{Q_{enclosed}}{\varepsilon_0}$	$\vec{\nabla}\cdot\vec{E} = \frac{\rho}{\varepsilon_0}$
Gauss' Law for Magnetism	$\oint\!\!\oint_S \vec{B}\cdot d\vec{A} = 0$	$\vec{\nabla}\cdot\vec{B} = 0$
Faraday's Law of Induction	$\oint_C \vec{E}\cdot d\vec{l} = -\frac{d\Phi_B}{dt}$	$\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t}$
Ampere-Maxwell Law	$\oint_C \vec{B}\cdot d\vec{l} = \mu_0 I_{enclosed} + \mu_0\varepsilon_0\frac{d\Phi_E}{dt}$	$\vec{\nabla}\times\vec{B} = \mu_0\vec{J} + \mu_0\varepsilon_0\frac{\partial\vec{E}}{\partial t}$

Differential Form of Gauss' Law

The **divergence theorem** tells us that if V is the volume enclosed by a closed surface S, then the total flux through S ("total outwards flow through the surface") is equal to the sum of all "small outward flows" (divergences) contained in V.

Mathematically:

$$\oint\!\!\oint_S \vec{F}\cdot d\vec{A} = \iiint_V \vec{\nabla}\cdot\vec{F}\,dV.$$

So, following Gauss' Law:

$$\oint\!\!\oint_S \vec{E}\cdot d\vec{A} = \iiint_V \vec{\nabla}\cdot\vec{E}\,dV = \frac{Q}{\varepsilon_0}$$

$$\frac{d}{dV}\iiint_V \vec{\nabla}\cdot\vec{E}\,dV = \frac{dQ}{dV}\frac{1}{\varepsilon_0}$$

$$\vec{\nabla}\cdot\vec{E} = \frac{\rho}{\varepsilon_0}.$$

Gauss' Law (Differential Form)

The divergence of the electric field, $\vec{\nabla}\cdot\vec{E}$, at any point is:

$$\vec{\nabla}\cdot\vec{E} = \frac{\rho}{\varepsilon_0}$$

where:

- ρ is volumetric charge density, charge per unit volume ($dq = \rho\,dV$)
- ε_0 is the permittivity of free space.

2 Electric Potential

2.1 Electric Potential Energy

Let $\vec{F}_e$ be the electrostatic force on a charge q moving from point A to point B. The **electric potential energy** U_e is defined as the work done by an external agent to move the charge from A to B against the electrostatic force:

$$U_e = W_{\text{external}} = -W_{\text{electrostatic}} = -\int \vec{F}_e \cdot d\vec{l}.$$

If we assume q is only affected by one other point charge Q, we can substitute in Coulomb's Law:

$$U_e = -\int k_e \frac{qQ}{r^2} \hat{r} \cdot d\vec{l}.$$

Realize that $\hat{r}$ and $d\vec{l}$ are always in the same direction (since we are moving along the line between the two charges). Thus, we can let $dr = \hat{r} \cdot d\vec{l}$:

$$U_e = -k_e qQ \int \frac{1}{r^2}\, dr.$$

Solving the integral:

$$\begin{aligned} U_e &= -k_e qQ \left[-\frac{1}{r}\right]_{r_A}^{r_B} \\ &= k_e qQ \left(\frac{1}{r_B} - \frac{1}{r_A}\right). \end{aligned}$$

If we let final point be infinitely far away from point A ($r_B \to \infty$), then the electric potential energy at point A is:

$$U_e = k_e q \frac{Q}{r_A}.$$

Thus, we arrive at the general formula for electric potential energy between two point charges:

Electric Potential Energy from a Point Charge

The electric potential energy of a point charge q a distance r from another point charge Q is:

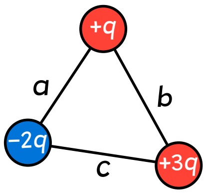
$$U_e = k_e \frac{qQ}{r}.$$

This is the amount of work available to move the charge q from distance r to an infinite distance away from Q.

- Just like all forms of energy, electric potential energy is a *scalar quantity* and is measured in *joules* (J).

- If both charges have the same sign, then U_e is positive, indicating that work wants to be done to separate the charges (they repel each other).
- If the charges have opposite signs, then U_e is negative, indicating that work is released when the charges come together (they attract each other).

Example: Electric Potential Energy of a Discrete System



A system of three point charges $+q$, $-2q$, and $+3q$ are arranged in a triangle with side lengths a, b, and c as shown. Calculate the total electric potential energy of the system.

The total potential energy in a system of discrete objects is the sum of potential energies between all pairs of objects.

Between $+q$ and $-2q$: $U_1 = k_e \dfrac{(+q)(-2q)}{a} = -2k_e \dfrac{q^2}{a}$.

Between $+q$ and $+3q$: $U_2 = k_e \dfrac{(+q)(+3q)}{b} = 3k_e \dfrac{q^2}{b}$.

Between $-2q$ and $+3q$: $U_3 = k_e \dfrac{(-2q)(+3q)}{c} = -6k_e \dfrac{q^2}{c}$.

Thus, the total electric potential energy of the system is:

$$U_e = U_1 + U_2 + U_3 = -2k_e \frac{q^2}{a} + 3k_e \frac{q^2}{b} - 6k_e \frac{q^2}{c} = k_e q^2 \left(-\frac{2}{a} + \frac{3}{b} - \frac{6}{c}\right).$$

Example: Electric Potential Energy of a Continuous System

A uniformly charged rod of length L and linear charge density λ lies along the x-axis from $x = 0$ to $x = L$. What is the electric potential energy of a point charge q located a distance d from the left end of the rod on the x-axis?

Since we are dealing with a continuous charge distribution, we use integration to find the total electric potential energy:

$$U_e = k_e q \int_{x=0}^{x=L} \frac{1}{r}\, dq$$

where r is the distance from the point charge to each small portion of charge on the rod.

Let x be how far from the left side of the rod we are (so we integrate from 0 to L). Then, $r = d + x$ and $dq = \lambda\, dx$:

$$U_e = k_e q \int_0^L \frac{1}{d+x} \lambda\, dx.$$

Since we know that the rod is *uniformly charged*, λ is constant, and we can now solve the integral and simplify:

$$\begin{aligned} U_e &= k_e q \lambda \int_0^L \frac{1}{d+x}\, dx \\ &= k_e q \lambda \ln(d+x)\Big|_{x=0}^{x=L} \\ &= k_e q \lambda (\ln(d+L) - \ln(d)) \\ &= k_e q \lambda \ln\left(\frac{d+L}{d}\right). \end{aligned}$$

Electric Potential Energy and Electrostatic Force

The electrostatic force is a **conservative force**. The work done by the force over a path only depends on the starting and ending positions, not the path taken.

The change in electric potential energy after following a path C is:

$$\Delta U_e = -\int_C \vec{F}_e \cdot d\vec{l}.$$

By the fundamental theorem of line integrals, we know that if a vector field $\vec{F}$ is the gradient of a scalar field f ($\vec{F} = \vec{\nabla} f$), then:

$$\int_C \vec{F} \cdot d\vec{l} = \int_C \vec{\nabla} f \cdot d\vec{l} = \Delta f.$$

This property only holds for conservative vector fields. Since the electrostatic force is conservative, we can apply this theorem:

$$\int_C \vec{\nabla} U_e \cdot \mathrm{d}l = \Delta U_e = \int_C -\vec{F}_e \cdot \mathrm{d}\vec{l}.$$

Here, we see that $\vec{\nabla} U_e = -\vec{F}_e$. This is the general definition of electric potential energy:

Electric Potential Energy

The electrostatic force $\vec{F}_e$ on a charge is equal to the negative gradient of its electric potential energy U_e:

$$\vec{F}_e = -\vec{\nabla} U_e.$$

In one dimension, this simplifies to:

$$F_{e_x} = -\frac{\mathrm{d}U_e}{\mathrm{d}x}.$$

2.2 Electric Potential

Electric Potential from a Point Charge

Let q be a charge in an electric field $\vec{E}$. The electric potential energy U_e of the charge is:

$$U_e = qV$$

where V is the **electric potential** at the position of the charge.

- Electric potential is the electric potential energy per unit charge ($V = U_e/q$).
  - This is similar to how electric field is defined as force per unit charge.
- The unit of electric potential is the *volt* (V).
  - A volt is defined as one joule per coulomb (V = J/C).
- The electric potential V due to a point charge Q at a distance r is:

$$V = \frac{U_e}{q} = k_e \frac{\mathrm{Q}}{\mathrm{r}}.$$

- Electric potential is a *scalar quantity*.
- Electric potential is often referred to as just "potential".
- Since $-\vec{\nabla} U_e = \vec{F}_e$, we can derive a relationship between electric potential and electric field:

$$\vec{F}_e = -\vec{\nabla}(qV) = -q\vec{\nabla}V \Longrightarrow \vec{E} = \frac{\vec{F}_e}{q} = -\vec{\nabla}V.$$

Electric Field from Electric Potential

The electric field $\vec{E}$ at any point is equal to the negative gradient of the electric potential V at that point:

$$\vec{E} = -\vec{\nabla}V.$$

In one dimension, this simplifies to:

$$E_x = -\frac{\mathrm{d}V}{\mathrm{d}x}.$$

2.3 Conservation of Electric Energy

3 Conductors and Capacitors

3.1 Electrostatics with Conductors

3.2 Redistribution of Charge Between Conductors

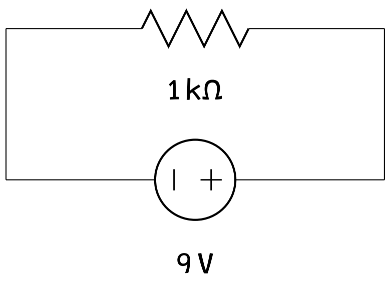
3.3 Capacitors

3.4 Dielectrics

4 Electric Current and Circuits

4.1 Electric Current

4.2 Electric Circuits



4.3 Resistance and Ohm's Law

4.4 Electric Power

4.5 Compound Direct Current Circuits

4.6 Kirchhoff's Loop Rule

4.7 Kirchhoff's Junction Rule

4.8 Resistor-Capacitor (RC) Circuits

An **RC circuit** is a circuit consisting of resistors and capacitors connected to a voltage source.

- The voltage source provides an electromotive force (emf) that drives current through the circuit.
- The resistors impede the flow of current, while the capacitors store and release energy.

By Kirchhoff's Loop Rule, the potential differences across the circuit must sum to zero:

$$\mathcal{E} - V_R - V_C = 0 \implies \mathcal{E} = V_R + V_C.$$

The potential difference across the resistor is given by Ohm's Law: $V_R = IR$, and the potential difference across the capacitor is given by $V_C = \frac{q}{C}$.

Thus, the equation for an RC circuit becomes:

$$\mathcal{E} = IR + \frac{q}{C}.$$

Since current I is the time derivative of charge $\frac{\mathrm{d}q}{\mathrm{d}t}$, we can rewrite the equation as:

$$\mathcal{E} = R\frac{\mathrm{d}q}{\mathrm{d}t} + \frac{1}{C}q.$$

This is a first-order linear differential equation. If we assume the capacitor is initially uncharged ($q(0) = 0$), we can solve for $q(t)$:

$$q(t) = C\mathcal{E}\left(1 - e^{-\frac{t}{RC}}\right).$$

5 Magnetic Fields and Electromagnetism

5.1 Magnetic Fields

- Magnetic fields are produced by moving charges (currents).
- Magnetic fields exert forces on moving charges.
- The magnetic field is represented by the vector $\vec{B}$, measured in *Teslas* (T).
  - A 1 Tesla magnetic field exerts a force of 1 Newton on a 1 Coulomb point charge moving at 1 m/s perpendicular to the field.
    $\mathrm{T} = \mathrm{N\,s\,C^{-1}\,m^{-1}} = \mathrm{kg\,s^{-2}\,A^{-1}}$ in SI base units.

5.2 Magnetism and Moving Charges

Magnetic Force

Let q be a charge moving with velocity $\vec{v}$ in a magnetic field $\vec{B}$. The **magnetic force** $\vec{F}_b$ on the charge is:

$$\vec{F}_b = q\vec{v} \times \vec{B}.$$

- The magnetic force is always perpendicular to both the velocity of the charge and the magnetic field.
- The magnetic force does no work on the charge, since it is always perpendicular to the velocity of the charge.
- If the charge is moving in a straight line (e.g. a straight wire), then the velocity is constant and we can write the magnetic force as:

$$\vec{F}_b = \frac{qL}{t} B \sin\theta = ILB \sin\theta.$$

  - Here, L is the length of a wire segment, and if it takes the charge q a time t to pass through the wire segment, then the current $I = \frac{q}{t}$.
  - The magnetic force on a charge moving through a wire segment is proportional to the length of that segment, the current through that segment, and the magnetic field strength.

Lorentz Force Law

The **Lorentz force** $\vec{F}$ on a charge q moving with velocity $\vec{v}$ in both electric and magnetic fields $\vec{E}$ and $\vec{B}$ is:

$$\vec{F} = \vec{F}_e + \vec{F}_b = q\vec{E} + q\vec{v} \times \vec{B} = q\left(\vec{E} + \vec{v} \times \vec{B}\right).$$

The Lorentz force describes how charged particles behave in electromagnetic fields.

5.3 Magnetic Fields of Current-Carrying Wires

Biot-Savart Law

Let I be the current through a wire segment $\mathrm{d}\vec{l}$. The magnetic field $\mathrm{d}\vec{B}$ at a point P due to this wire segment is:

$$\mathrm{d}\vec{B} = \frac{\mu_0}{4\pi}\frac{I\,\mathrm{d}\vec{l} \times \hat{r}}{r^2}$$

where:

- μ_0 is the **permeability of free space**
- $\hat{r}$ is the unit vector from the wire segment to point P
- r is the distance from the wire segment to point P

If a wire is curved in a path C, then the total magnetic field at point P is:

$$\vec{B} = \frac{\mu_0}{4\pi}\int_C \frac{I\,\mathrm{d}\vec{l} \times \hat{r}}{r^2}.$$

By the definition of the cross product, the magnitude of the magnetic field contribution from the wire segment is:

$$\left|\mathrm{d}\vec{B}\right| = \mathrm{d}B = \frac{\mu_0}{4\pi}\frac{I\,\mathrm{d}l\sin\theta}{r^2}$$

where θ is the angle between $\mathrm{d}\vec{l}$ and $\hat{r}$.

Example: Magnetic Field from a Straight Current-Carrying Wire

A long, straight wire carries a current I. What is the magnetic field $\vec{B}$ at a point P a distance r away from the wire?

A long, straight wire carries a current I. What is the magnetic field $\vec{B}$ at a point P a distance r away from the wire?

Consider a small wire segment $\mathrm{d}l$ at an angle θ from the horizontal axis. The distance from this wire segment to point P is $\frac{r}{\cos\theta}$. Thus, by the Biot-Savart Law:

$$\mathrm{d}B = \frac{\mu_0}{4\pi}\frac{I\,\mathrm{d}l\sin\theta}{\left(\frac{r}{\cos\theta}\right)^2} = \frac{\mu_0 I}{4\pi r^2}\,\mathrm{d}l\sin\theta\cos^2\theta.$$

5.4 Ampere’s Law

Ampere's Law

$$\oint_C \vec{B} \cdot \mathrm{d}\vec{l} = \mu_0 I_{\text{enclosed}}$$

6 Electromagnetic Induction

6.1 Magnetic Flux

6.2 Electromagnetic Induction

6.3 Induced Currents and Magnetic Forces

6.4 Inductance

6.5 Circuits with Resistors and Inductors (LR)

6.6 Circuits with Capacitors and Inductors (LC)

6.7 Circuits with Resistors, Inductors, and Capacitors (RLC)

$$L\frac{\mathrm{d}^2 q}{\mathrm{d}t^2} + R\frac{\mathrm{d}q}{\mathrm{d}t} + \frac{1}{C}q = \mathcal{E}$$